

QUANTICO MCAF, VA, USA

WMO: 724035

Lat: 38.504N

Lon: 77.305W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13773

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.9	-19.1	0.7	-6.1	-16.5	0.9	-4.3	11.3	2.5	10.1	3.4	3.6	290	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	33.6	24.6	32.3	24.4	31.1	23.9	26.6	31.3	25.7	30.3	25.0	29.4	3.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.3	29.4	24.3	19.2	28.4	23.6	18.5	27.7	82.6	31.2	79.0	30.5	75.9	29.5	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.1	7.2	-12.8	36.2	3.3	1.8	-15.2	37.6	-17.2	38.6	-19.0	39.6	-21.5	41.0		
			-13.7	28.3	3.0	1.4	-15.9	29.3	-17.7	30.1	-19.4	30.9	-21.6	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	2.1	3.7	7.9	14.0	18.8	23.6	25.9	25.0	21.4	15.0	8.8	4.4	(6)
		DBStd	9.34	5.32	5.03	5.13	4.47	4.09	3.12	2.63	2.59	3.44	4.26	4.37	4.58	
		HDD10.0	816	252	186	103	12	0	0	0	0	7	73	183		
		HDD18.3	2283	504	411	324	144	47	3	0	0	122	286	432		
		CDD10.0	2374	6	8	39	132	272	407	492	466	343	161	38	11	
		CDD18.3	798	0	0	2	13	60	160	234	208	103	17	1	0	
		CDH23.3	6669	1	1	16	101	397	1306	2253	1825	684	82	3	1	
		CDH26.7	2248	0	0	2	23	94	426	882	634	174	12	0	0	
(14)	Wind	WSAvg	3.3	3.3	3.4	3.9	3.8	3.3	3.0	2.9	2.9	3.2	3.2	3.3	3.1	(14)
(15)	Precipitation	PrecAvg	968	73	75	86	70	92	81	81	95	92	66	82	80	(15)
(16)		PrecMax	1337	146	209	154	146	234	258	182	217	343	165	203	197	(16)
(17)		PrecMin	544	34	9	37	6	14	17	20	13	3	1	11	6	(17)
(18)		PrecStd	209	24	47	33	36	54	60	41	53	76	41	63	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	18.9	21.1	25.9	29.7	31.6	34.2	36.4	34.8	32.9	28.0	23.0	20.0		(19)
(20)		0.4% MCWB	14.0	14.6	15.8	18.3	22.3	24.2	25.7	25.2	24.0	22.2	16.2	15.0		(20)
(21)		2% DB	15.2	16.7	21.6	26.3	29.0	32.2	34.0	33.0	30.4	25.6	19.8	16.4		(21)
(22)		2% MCWB	11.3	11.6	14.0	17.4	21.7	23.8	25.1	24.8	23.4	20.5	14.7	12.8		(22)
(23)		5% DB	12.1	13.4	18.8	23.6	27.4	30.7	32.5	31.7	28.6	23.5	17.6	13.5		(23)
(24)		5% MCWB	9.3	8.8	12.9	16.6	21.0	23.5	24.8	24.3	22.4	19.5	13.6	10.7		(24)
(25)		10% DB	9.4	11.1	16.1	21.4	25.7	29.1	31.2	30.2	27.1	22.0	15.9	11.2		(25)
(26)		10% MCWB	6.8	7.4	10.8	15.5	19.8	22.8	24.2	23.6	21.7	18.2	12.4	8.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	14.9	15.8	17.5	21.3	24.7	26.7	28.0	27.7	26.0	23.9	18.5	16.5		(27)
(28)		0.4% MCDB	18.1	19.2	22.5	24.7	28.9	31.4	32.5	31.6	29.7	26.2	20.4	18.9		(28)
(29)		2% WB	12.2	12.5	15.5	19.1	23.1	25.5	26.7	26.4	24.7	22.3	16.5	13.8		(29)
(30)		2% MCDB	14.2	15.5	20.5	23.7	27.3	30.2	31.6	30.9	28.4	24.5	18.7	15.9		(30)
(31)		5% WB	9.7	9.9	13.7	17.8	22.0	24.7	25.8	25.4	23.9	20.6	14.7	11.5		(31)
(32)		5% MCDB	11.6	12.5	17.9	22.1	26.0	29.0	30.8	29.9	27.3	22.9	16.9	13.1		(32)
(33)		10% WB	7.2	7.6	11.5	16.5	20.9	23.9	25.1	24.7	22.9	18.7	13.0	8.8		(33)
(34)		10% MCDB	9.0	10.5	15.2	20.8	24.5	27.9	29.8	28.6	25.7	21.2	15.1	10.6		(34)
(35)	Mean Daily Temperature Range	MDBR	8.4	9.3	9.9	10.7	9.7	9.2	9.0	8.9	9.1	9.9	9.7	8.5		(35)
(36)		5% DB MCDBR	12.2	13.3	13.9	13.6	11.5	10.7	10.3	10.2	10.3	11.2	12.1	11.7		(36)
(37)		5% DB MCWBR	9.2	9.2	8.1	7.2	6.3	5.5	4.8	4.6	5.1	6.8	8.3	9.1		(37)
(38)		5% WB MCDBR	11.1	12.1	12.5	11.7	9.7	9.4	9.5	9.1	8.6	9.1	10.0	10.7		(38)
(39)		5% WB MCWBR	9.3	9.3	8.4	7.3	5.7	5.3	4.9	4.7	4.9	6.2	8.0	9.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.321	0.356	0.401	0.440	0.497	0.518	0.515	0.425	0.360	0.324	0.310		(40)
(41)		taud	2.502	2.442	2.401	2.292	2.214	2.093	2.058	2.045	2.278	2.460	2.519	2.528		(41)
(42)		Ebn at Noon	876	906	904	877	846	795	774	764	820	850	847	846		(42)
(43)		Edh at Noon	78	94	109	129	142	160	165	163	122	92	77	72		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.15	2.97	3.96	4.93	5.59	6.15	5.99	5.36	4.38	3.33	2.44	1.89		(44)
(45)		RadStd	0.19	0.33	0.28	0.36	0.60	0.46	0.31	0.27	0.51	0.37	0.20	0.18		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (74 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+114
		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	+79

Nomenclature: See separate page